

Single Organ Segmentation (Model DeepLabv3+)

(Intestinal Veins, Ureter, Inferior mesenteric artery (IMA), Vesicular glands (Vglands), Pancreas)

(Framework=PyTorch, Treshold=0.40, trainig size=0.8, vaidation size=0.1, test_size=0.1)

a) **Intestinal Veins** (Train image=**853**, validation image=**106**, Test image =**108**)

image size 128x128 (Model DeepLabv3+)

Number of Epochs	10 Epoch	30 Epoch	50 Epoch
IoU	0.4588	0.5325	0.5450
Precision	0.5200	0.5637	0.5940
Recall	0.8297	0.8795	0.8916
Specificity	0.9908	0.9932	0.9925
F1 Skore	0.6094	0.6713	0.6920
Running Time (seconds)	394.69	1140.49	1937.87

image size 256x256 (Model DeepLabv3+)

Number of Epochs	10 Epoch	30 Epoch	50 Epoch
IoU	0.4741	0.5497	0.5606
Precision	0.5076	0.5815	0.6098
Recall	0.8551	0.9176	0.8648
Specificity	0.9912	0.9929	0.9944
F1 Skore	0.6239	0.6946	0.6993
Running Time (seconds)	437.51	1182.47	1846.08

b) **Ureter** (Train image=**1020**, Test image=**128**, validation image=**127**)

image size 128x128 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4543	0.4886	0.5254
Precision	0.6590	0.6439	0.6178
Recall	0.5777	0.6286	0.7587
Specificity	0.9979	0.9969	0.9953
F1 Skore	0.5764	0.6022	0.6553
Running Time (seconds)	445.36	1331.67	2027.37

image size 256x256 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4850	0.5107	0.5722
Precision	0.6139	0.6303	0.6475
Recall	0.6767	0.7136	0.8077
Specificity	0.9959	0.9957	0.9957
F1 Skore	0.6120	0.6361	0.6978
Running Time (seconds)	422.97	1216.44	1913.93

c) **Inferior mesenteric artery** (Train image=872, Test image=109, validation image=109)

image size 128x128 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.5159	0.5863	0.5925
Precision	0.5698	0.7162	0.6855
Recall	0.8466	0.7794	0.8149
Specificity	0.9838	0.9927	0.9915
F1 Skore	0.6551	0.7154	0.7186
Running Time (seconds)	394.05	1633.66	1864.78

image size 256x256 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.5465	0.6101	0.6747
Precision	0.7205	0.7290	0.7734
Recall	0.7023	0.7877	0.8437
Specificity	0.9935	0.9934	0.9950
F1 Skore	0.6813	0.7341	0.7894
Running Time (seconds)	455.42	1542.32	1834.62

d) **Vesicular Glands** (Train image=958, Test image=121, validation image=119)

image size 128x128 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4593	0.5224	0.5365
Precision	0.5430	0.6610	0.6515
Recall	0.7373	0.6744	0.7317
Specificity	0.9859	0.9924	0.9916
F1 Skore	0.5890	0.6453	0.6609
Running Time (seconds)	449.99	1090.24	1803.17

image size 256x256 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.5149	0.5772	0.6048
Precision	0.6199	0.6620	0.7179
Recall	0.7481	0.7900	0.7714
Specificity	0.9885	0.9927	0.9931
F1 Skore	0.6411	0.6950	0.7205
Running Time (seconds)	383.05	1162.68	1909.50

e) **Pancreas** (Train image=938, Test image=118, validation image=117)

image size 128x128 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4730	0.5250	0.5532
Precision	0.5691	0.6131	0.6453
Recall	0.6985	0.7522	0.7775
Specificity	0.9868	0.9910	0.9907
F1 Skore	0.5962	0.6515	0.6888
Running Time (seconds)	468.61	1333.59	1866.52

image size 256x256 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.4934	0.5342	0.6183
Precision	0.7272	0.6915	0.7142
Recall	0.5561	0.6671	0.8079
Specificity	0.9968	0.9940	0.9939
F1 Skore	0.6081	0.6546	0.7427
Running Time (seconds)	410.06	1175.77	1921.20

image size 512x512 (Model DeepLabv3+)

Number of Epochs	10	30	50
IoU	0.6081	0.6679	0.6761
Precision	0.7339	0.7919	0.7650
Recall	0.7640	0.7897	0.8460
Specificity	0.9942	0.9968	0.9954
F1 Skore	0.7271	0.7763	0.7891
Running Time (seconds)	587.47	2365.56	5862.49